

可视化研发过程开发和管理解决方案



北京经纬恒润科技有限公司
2019年11月

议程 Agenda

- 研发过程管理概述

An Engineering Process Management Overview

- 基于模型的研发过程管理平台Stages

Stages: a Model-based Engineering Process Management Platform

- 客户案例

Customer Stories

- 总结

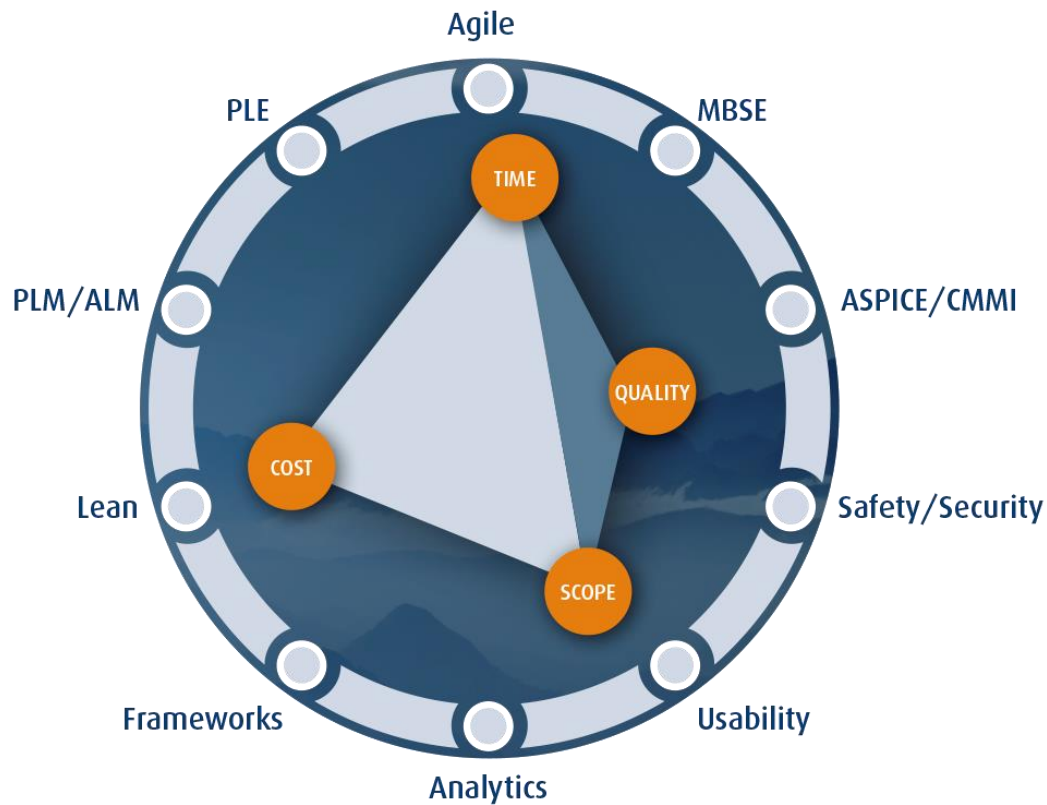
Summary

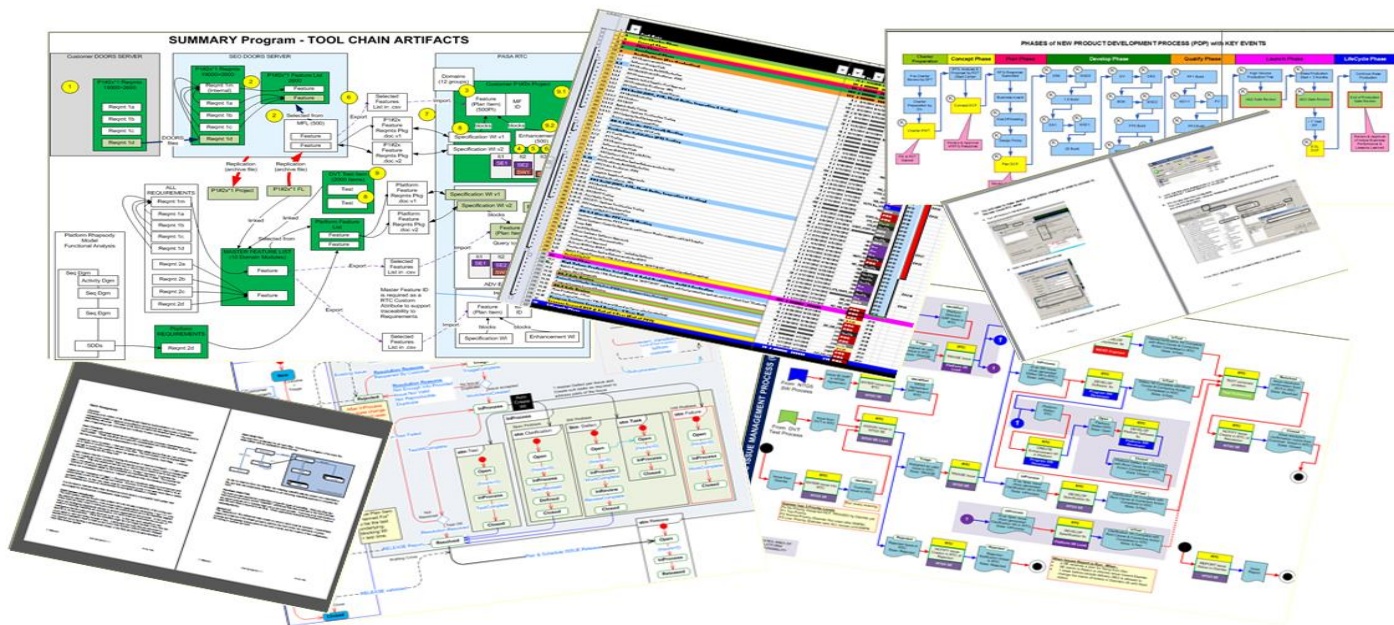
PART 1

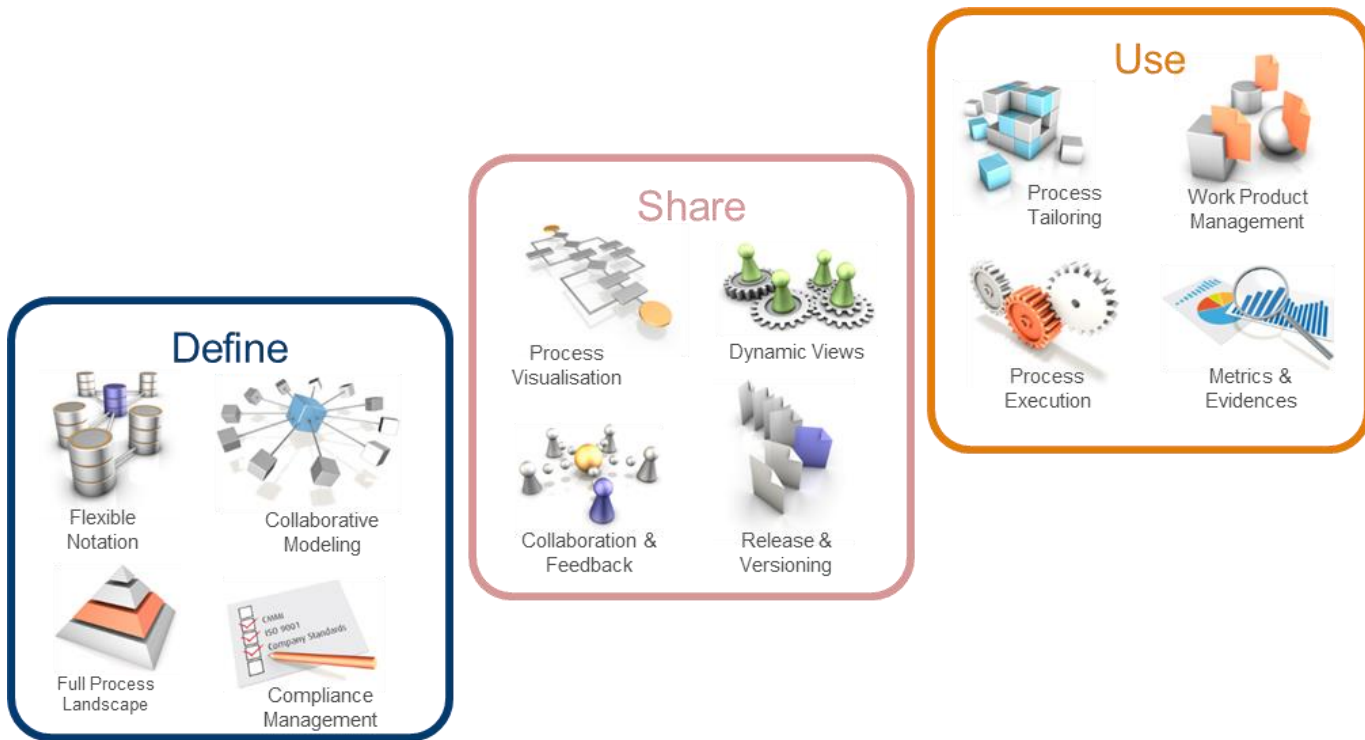
研发过程管理概述

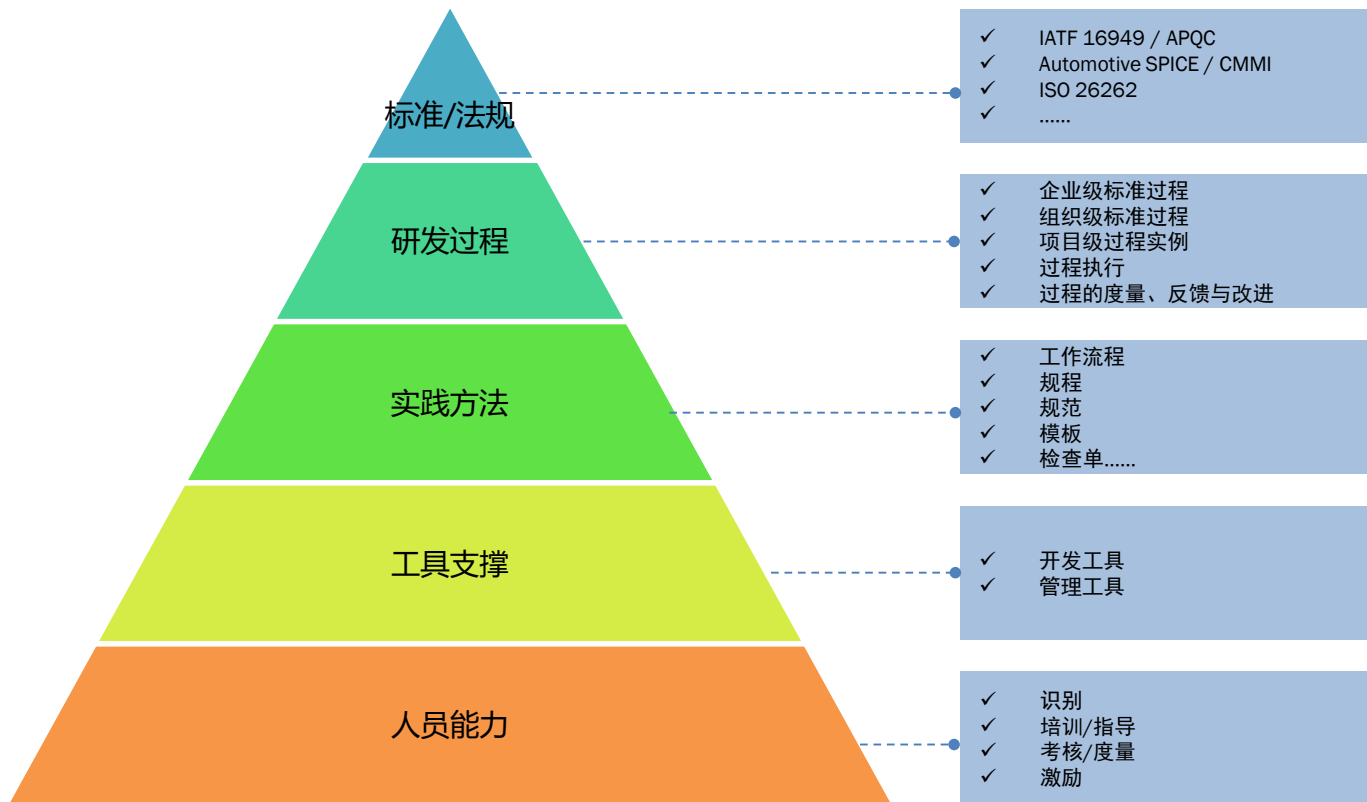
An Engineering Process Management Overview

研发过程领域的关键词与新趋势



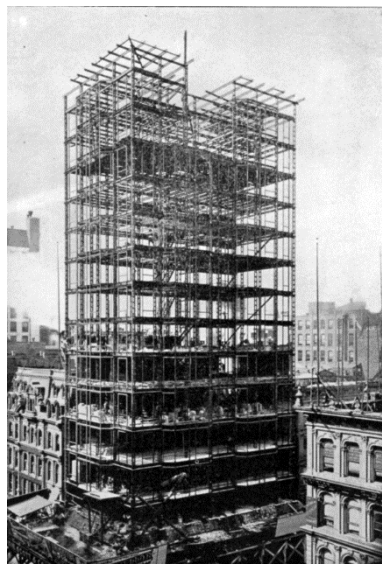






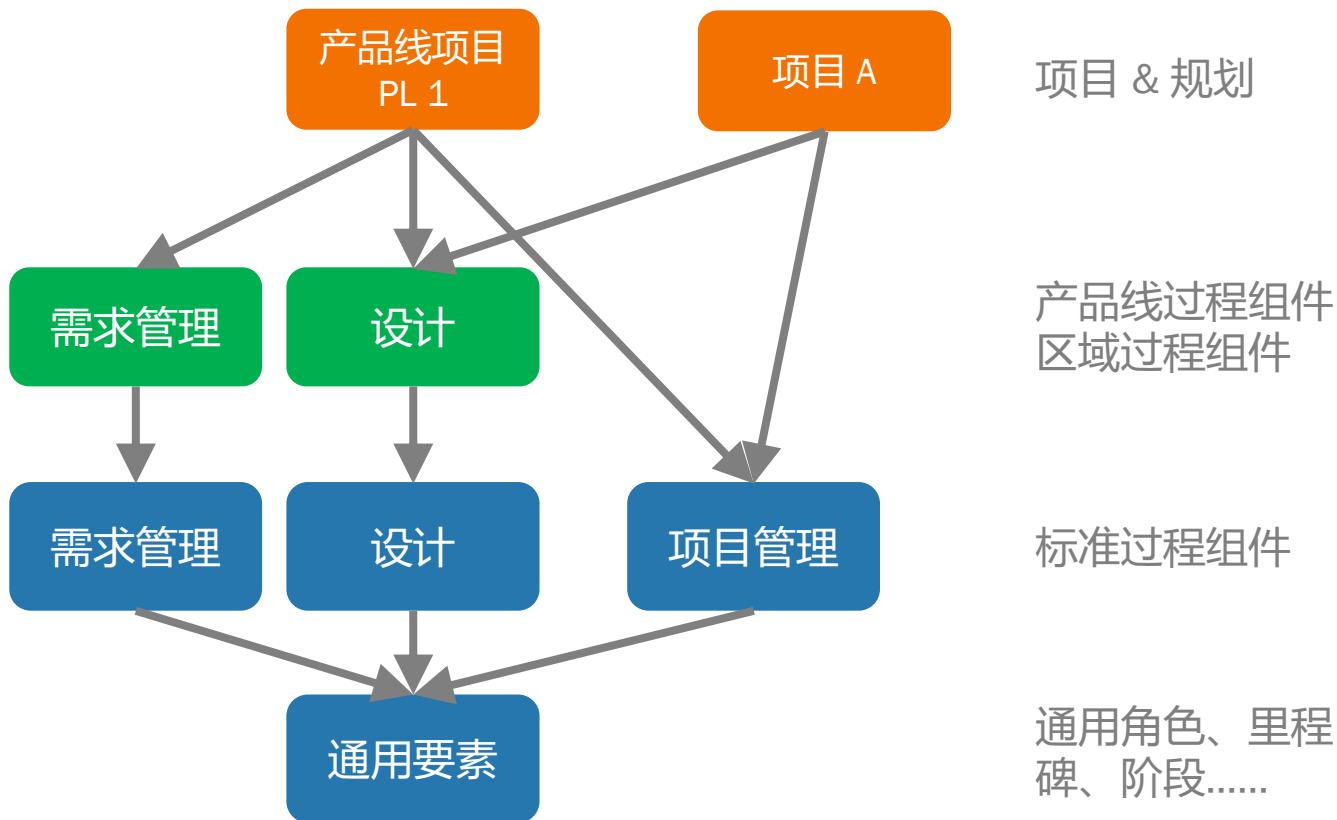
研发过程管理需要关注过程的宏观和微观架构

Intra Process Structure 微观过程结构

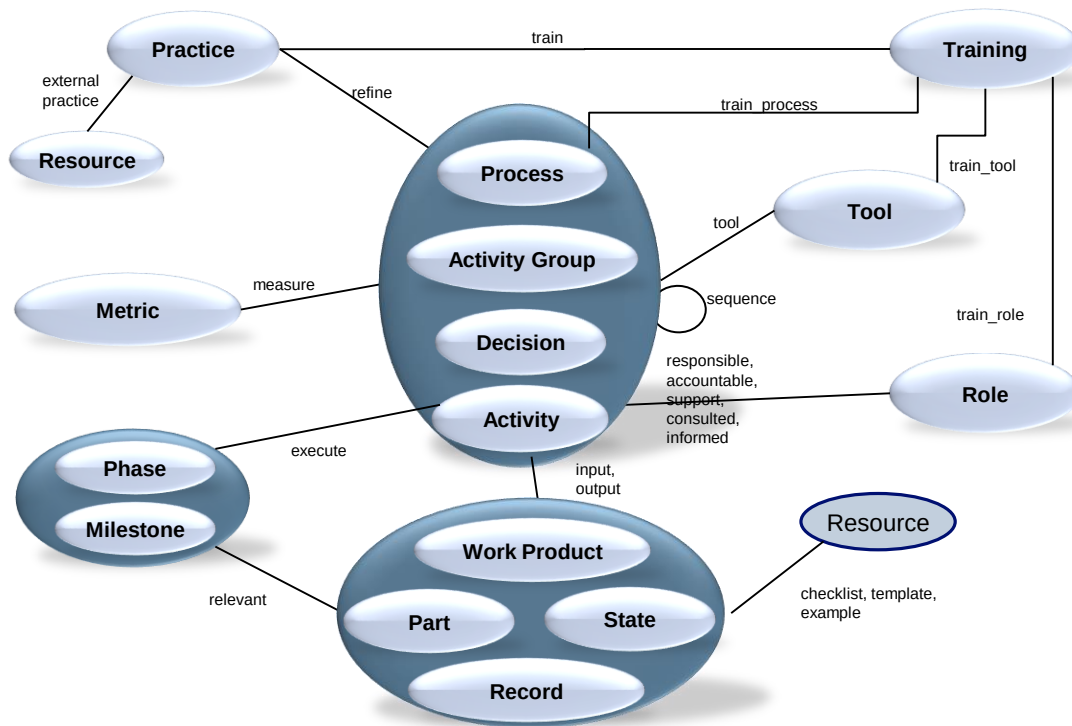


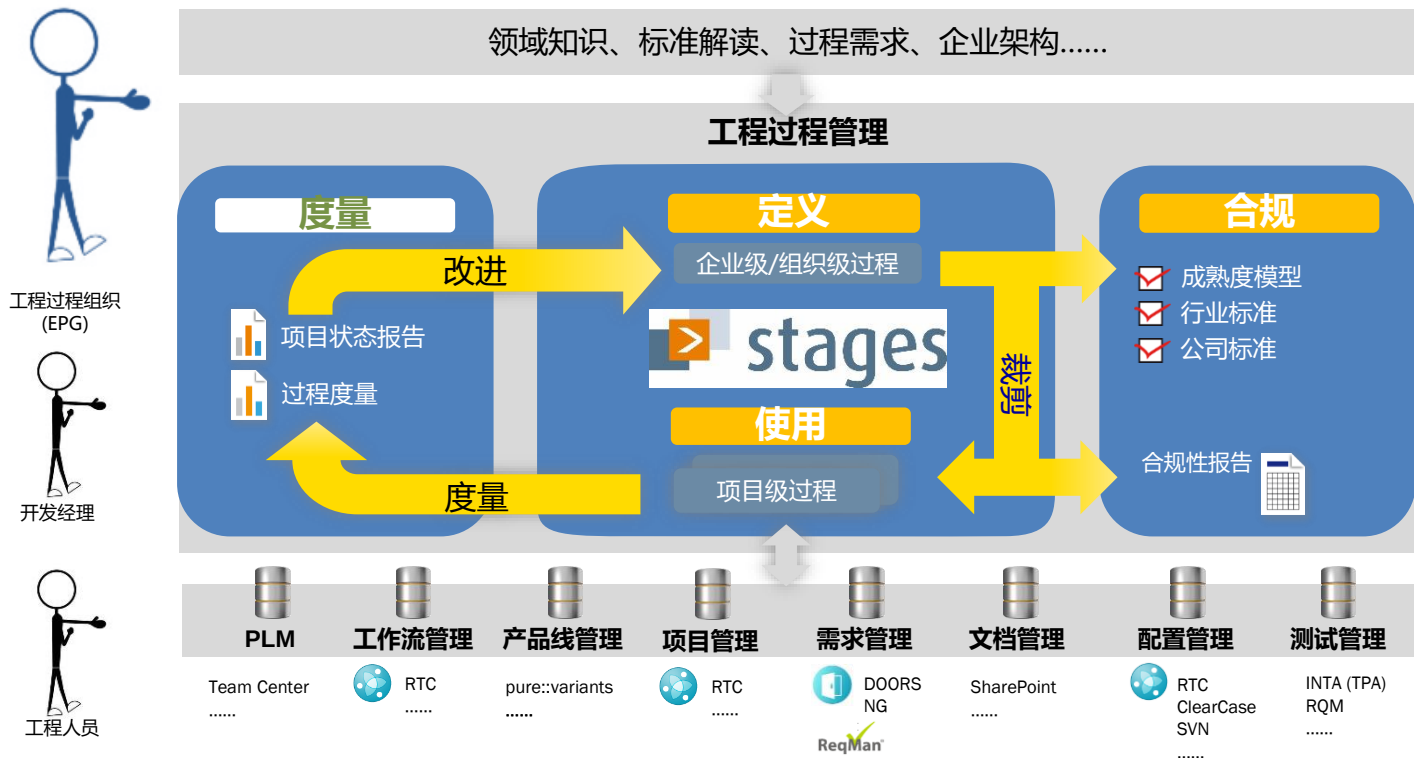
Inter Process Structure 宏观过程结构

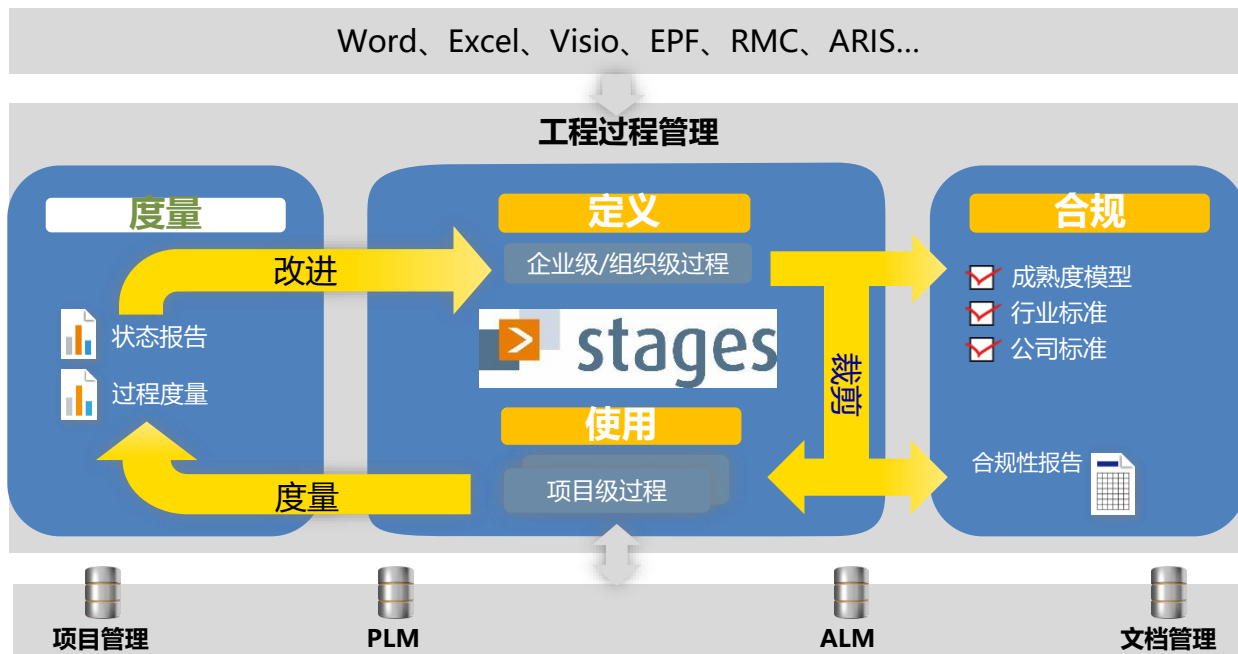


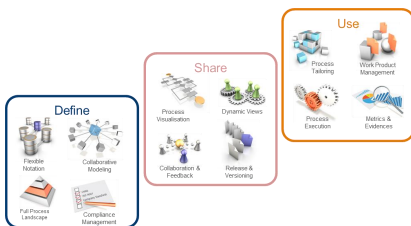
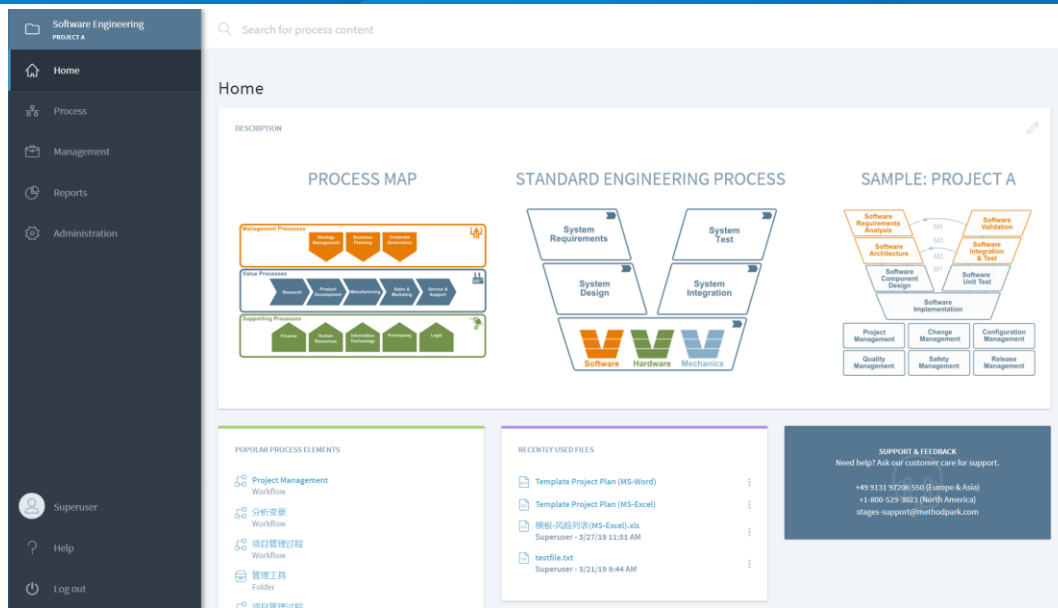


■ 一个过程中包含的内容 Elements in process









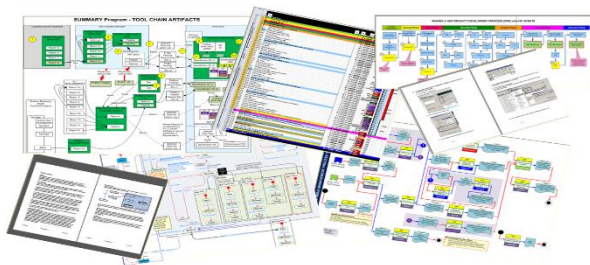
The Process Navigation System



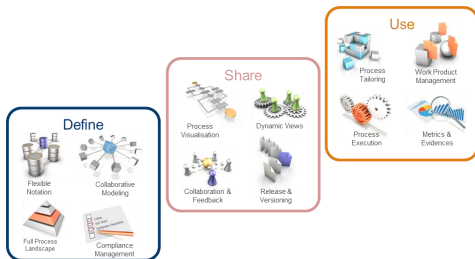
PART 2

基于模型的研发过程管理平台Stages

Stages: a Model-based Engineering Process Management Platform



- 将文档化的过程体系转变为基于模型的过程体系模型，增强过程的一致性和被组织接受的程度，有效降低过程合规（ex: ASPICE）的复杂性



- 将过程模型在组织内分享，并与过程实际执行有机结合

Software Engineering
PRODUCT DEVELOPMENT

Search for process content

Process - Workflows and Activities

SEEng V1.2

Project Management

WORKFLOW

Perform Feasibility Study
ACTIVITY

Create/Update Risk Analysis
ACTIVITY

Create/Update Software Project Plan
ACTIVITY

Review Software Project Plan
ACTIVITY

Create/Update Status Report
ACTIVITY

SEQUENCE

Predecessor

Perform Feasibility Study

Successor

Create/Update Software Project Plan

PHASES

Phases

Idea

DESCRIPTION

Risk Analysis should be performed preventive and periodically in regular time intervals which should be as short as possible. The results shall be documented in the will include the following steps:

- Identifying and assessing risks and planning measures
- Monitoring risks and keeping track of the effectiveness of the measures.

WORK PRODUCTS

Inputs

Feasibility Study draft

Outputs

Feasibility Study released

Risk Analysis

GUIDANCE SUPPLEMENTS

Tools

ROLES

Responsible

Risk Manager

Accountable

Supporting

Quality Engineer

Security Officer

System Analyst

INFORMED

ATTRIBUTES

None

Tailoring Assistant

STEP THROUGH THE TAILORING QUESTIONS

Back

Does the project have safety relevant aspects?

Yes

No

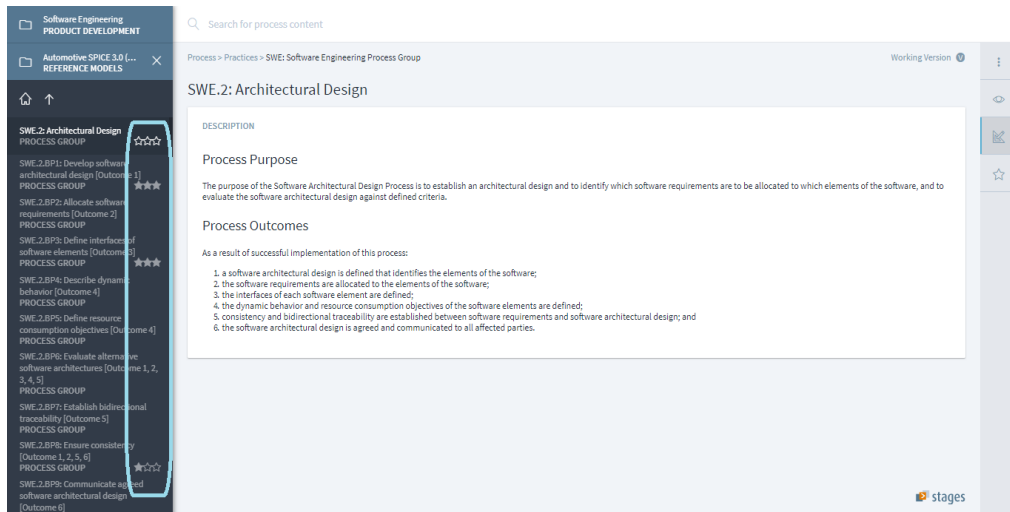
I don't know

Cancel

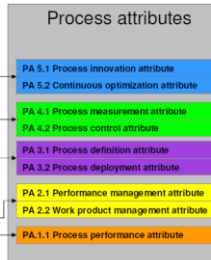
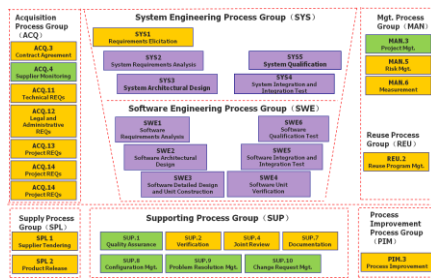
Preview Tailoring

Execute

- 提供结构化的过程裁剪功能，有效应对ASPICE CL3对组织级过程在项目上实例化的要求，由开发经理（项目经理）配置成适配项目的过程



将参考模型(PRM)或标准（如ASPIICE、ISO 26262）条目化地构建在Stages系统内
质量专家能够在Stages中维护组织自身的研发过程与PRM的差距，可以生成用于过程评估/审计用的证据



Process Area name	Process Area Code	过程域 (中文)	Category 类别	Maturity Level 成熟度等级	Capability Level 能力等级1	Capability Level 能力等级2	Capability Level 能力等级3
Configuration Management	CM	配置管理	Support	2			
Measurement and Analysis	MA	度量与分析	Support	2			
Project Monitoring and Control	PWC	项目监控	Process Management	2			
Project Planning	PP	项目计划	Support	2			
Process and Product Quality Assurance	PPQA	过程和产品质量保证	Support	2			
Requirements Management	REQM	需求管理	Process Management	2			
Supplier Agreement Management	SAM	供应商管理	Process Management	2			
Decision Analysis and Resolution	DAR	决策分析与决议	Support	2			
Integrated Project Management	IPM	集成项目管理	Process Management	2			
Organizational Process Definition	OPD	组织过程定义	Process Management	2			
Organizational Process Focus	OPF	组织过程焦点	Process Management	2			
Organizational Training	OT	组织培训	Process Management	2			
Product Integration	PI	产品集成	Engineering	2			
Requirements Development	RD	需求开发	Process Management	2			
Risk Management	RSM	风险管理	Process Management	2			
Technical Solution	TS	技术方案	Engineering	2			
Validation	VAL	验证	Engineering	2			
Verification	VER	验证	Engineering	2			
Organizational Process Performance	OPP	组织过程绩效	Process Management	4			
Quantitative Project Management	QPM	量化项目管理	Process Management	4			
Panel Analysis and Resolution	PAN	问题分析与决议	Support	2			
Organizational Performance Management	OPM	组织绩效管理	Process Management	4			

The image displays three screenshots of the 'Software Engineering PRODUCT DEVELOPMENT' interface, illustrating the process of tailoring project workflows.

Top Left Screenshot: Tailoring Assistant
This window shows a 'Tailoring Assistant' with a 'STEP THROUGH THE TAILORING QUESTIONS' section. A red box highlights the question 'Does the project have safety relevant aspects?' with three possible answers: 'Yes', 'No', and 'I don't know'. At the bottom, there are 'Cancel', 'Preview Tailoring', and 'Execute' buttons.

Top Right Screenshot: Tailoring Results
This window shows the results of the tailoring process. It includes a search bar, a question 'Does the project have safety relevant aspects?' with a 'No' answer, and a 'JUSTIFICATION' section. Below this, a list of 'REMOVE, IF SELECTED' items is shown, including 'Workflows and Activities' (with a red box around 'Perform Hazard Analysis'), 'Work Products', and 'Roles' (with 'Safety Engineer' listed).

Bottom Screenshot: Workflows and Activities
This window shows a list of 'Workflows and Activities' under the 'Process' tab. A red box highlights the following items: 'Change Management', 'Configuration Management', 'Project Management', 'Quality Management', 'Requirements Management', 'System & Software Design', and 'Verification & Validation'.

- 提供结构化的过程裁剪功能，有效应对ASPICE CL3对组织级过程在项目上实例化的要求，由开发经理（项目经理）配置成适配项目的过程

Project Plan Export

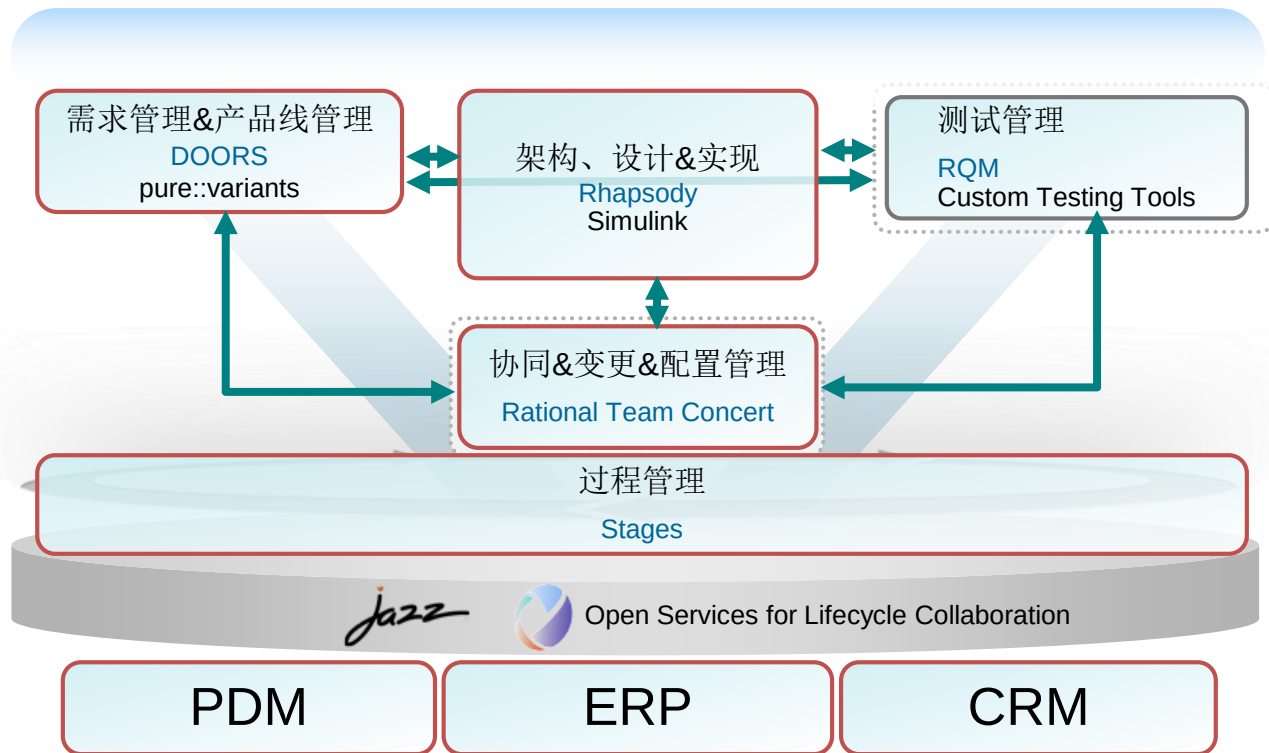
REPORT

Run

Internal ID	Activity Name	Duration	Predecessor IDs	Level	Responsible Role	Supporting Role	Inputs	Outputs	URL
34429	Project Management	1		0	Project Manager				http://localhost:8081/Yw
34461	Perform Feasibility Study	1		1	Risk Manager	System Analyst;System Architect	Software Requirements Specification	Feasibility Study	http://localhost:8081/
50105	新活动	1	34461	1			Status Report	评审报告	http://localhost:8081/
34468	Create/Update Risk Analysis	1	34461	1	Risk Manager	Quality Engineer;Security Officer;System Analyst	Feasibility Study	Feasibility Study;Risk Analysis	http://localhost:8081/
34473	Create/Update Software Project Plan	1	34468	1	Project Manager	Change Manager;Quality Manager	Feasibility Study;Risk Analysis	Software Project Plan;Milestone Schedule	http://localhost:8081/
34427	Review Software Project Plan	1	34473	1	Project Manager	Quality Manager;Risk Manager	Software Project Plan	Software Project Plan	http://localhost:8081/
34465	Create/Update Status Report	1	34427	1	Project Manager	Change Manager;Configuratic Manager;Quality Engineer	Software Project Plan;Milestone Schedule	Status Report	http://localhost:8081/
34463	Review Status Report	1	34465	1	Quality Manager		Status Report	Status Report	http://localhost:8081/

May 14, 2019, 4:55 AM

- 可以将裁剪后的项目过程发布成WBS，被下游项目管理系统使用



The screenshot displays the Jazz Administration web application. The top navigation bar includes menus for File, Edit, Navigate, Search, Project, Run, Window, and Help. The left sidebar shows a tree view of project and team areas, including Repository Connections, Stages for GA, Stages - Process Management, Builds, Enterprise Extensions, Plans, Reports, Source Control, Work Items, Favorites, Feeds, My Repository Workspaces, My Team Areas, and Work Item History.

The main content area is titled "System Design" and shows a process flow diagram with three steps: "System - System Design", "System Design", and "System Test". Below the diagram, there is a section for "FILES AND LINKS" with a table of items:

NAME	FILE NAME	LAST CHANGE	STATUS
system design.doc	system design.doc	2012/08/08 PM	Draft

Below the table, there is a "Drag & Drop" area with the text "your files here, or remove".

The bottom section, titled "Work Items", shows a table of recently created work items:

ID	Status	P	S	Summary	Owned By	Created By
180	New			Establish Risk Analysis	Unassigned	Erich Meier
179	New			CR-177: Perform Safety Analysis	Unassigned	Erich Meier
178	New			CR-177: Perform System Design	Unassigned	Erich Meier
177	Approved				Unassigned	Erich Meier

The bottom status bar shows a search for ID or Text, a link to "Stages - Process Ma...", and a URL: "http://localhost:8080/p...&type=ArtifactInstance".

The image displays three overlapping screenshots of a software interface for 'Risk Analysis'.

Top Screenshot (Risk Analysis): Shows a process flow diagram with 'Project Management' and 'Risk Analysis' steps. Below, a table lists files and links.

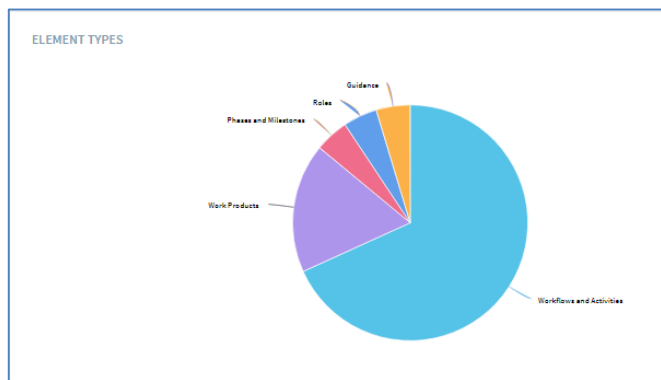
NAME	AUTHOR	LAST CHANGE	STATE
AE projectA Risk Analyse.docx	Superuser	5/22/19 5:42 PM	Draft

Middle Screenshot (Risk Analysis): Similar to the top one, but with a context menu open for the file 'AE projectA Risk Analyse.docx'.

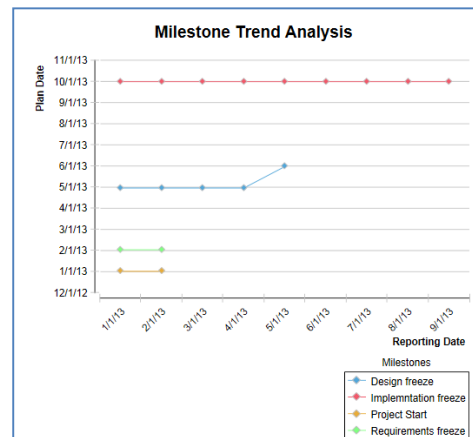
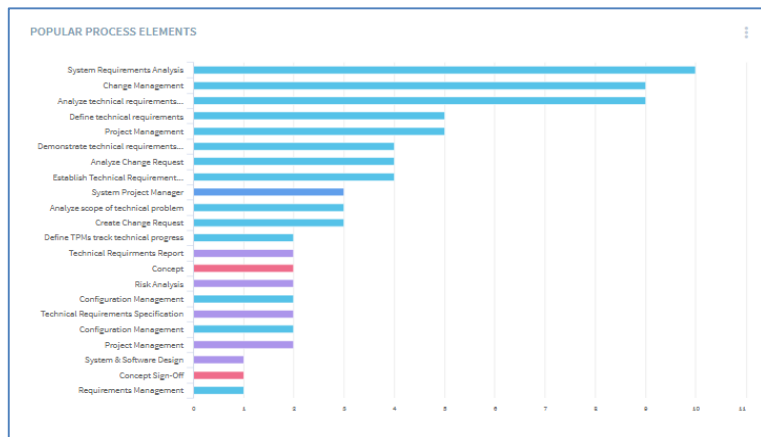
- Download & lock
- Upload
- Lock
- Set state
- Version History
- Refresh
- Rename
- Properties
- Delete

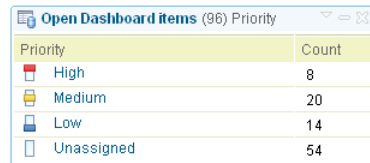
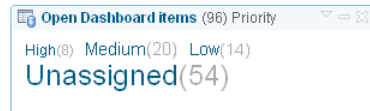
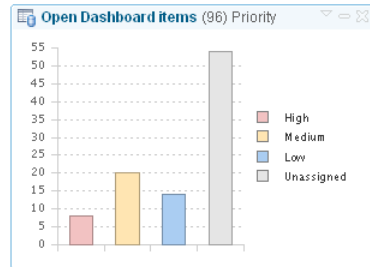
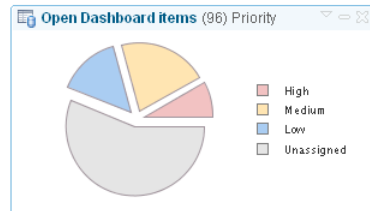
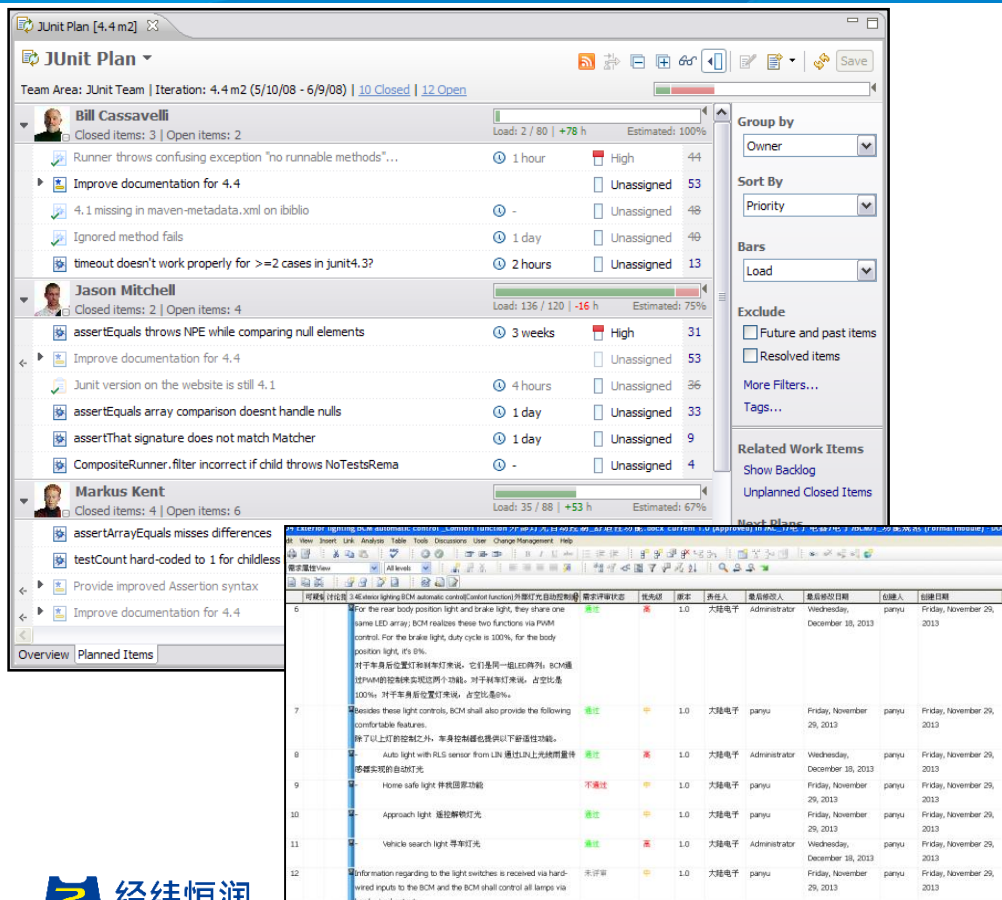
Bottom Screenshot (Version History: AE projectA Risk Analyse.docx): Shows a table of version history.

MODIFIED BY	DATE	VERSION	STATE	MORE DETAILS
kxy	5/22/19 5:42 PM	11	Draft	1
pm	5/22/19 5:41 PM	10	Review	1
Superuser	5/22/19 5:39 PM	9	Draft	1



Process Area	Process	Phases			Processes & Activities			Work Products			Roles			Metrics			Practices		
		ETP	%		ETP	%		ETP	%		ETP	%		ETP	%		ETP	%	
Change Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Configuration Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Electronics Design	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OTHER	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Project Management	Decision Management	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Project Assessment and Control	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Project Planning	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Risk Management	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Quality Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Release Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Requirements Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Safety Management	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Stakeholder Requirements Definition	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Supporting Activities	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Architectural Design	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Implementation	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Integration	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Requirements Analysis	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Validation	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
System Verification	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Verification & Validation	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Component Test	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Functional Test	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Integration Test	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	OTHER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Software Validation	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	System Test	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0





PART 3

客户案例

Customer Stories

Ω 问题:

- ◆ 结构、硬件、软件专业都是独立的工程过程，沟通不畅
- ◆ 超过400个不同的工具和库，冗余且难以维护一致性
- ◆ 项目效率低下

Ω 需求:

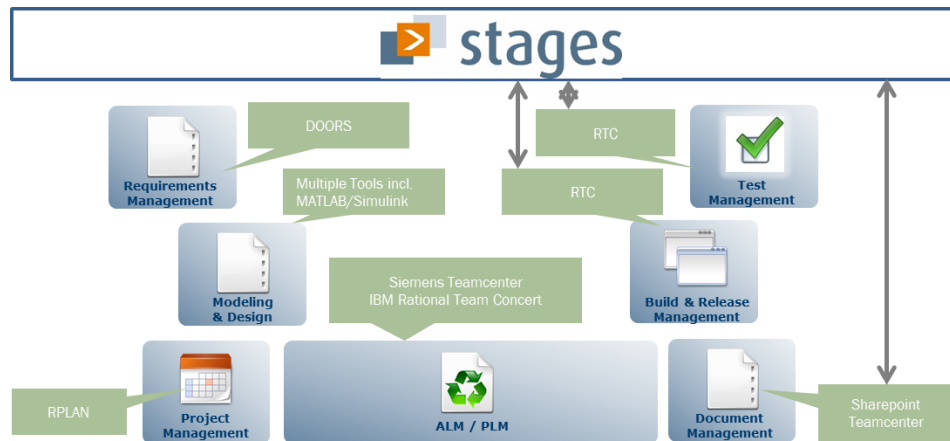
- ◆ 搭建新的可视化过程管理平台，建立产品线工程能力
- ◆ 创造以安全为信条的企业文化

Ω 解决方案:

- ◆ 2010年开始实施Stages
- ◆ 约3500个license

Ω 收益:

- ◆ 提升了工程师对过程的接受度
- ◆ 降低了70%的工具/库的不兼容性



通过Stages作为“接口”，连通项目管理、需求管理、测试管理、构建和发布管理（ALM）、PLM、文档管理能力，形成有机整体平台方案

Ω 问题:

- ◆ 独立业务单元众多，且由于历史原因，流程、文化、管理方式多样
- ◆ 各专业过程独立，协作效率低下
- ◆ 项目进度和状态不透明
- ◆ 超过万人的工程团队，且拥有众多工具和库

Ω 需求:

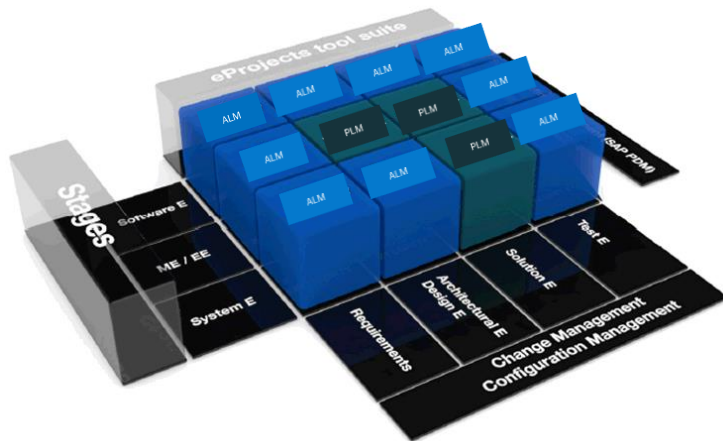
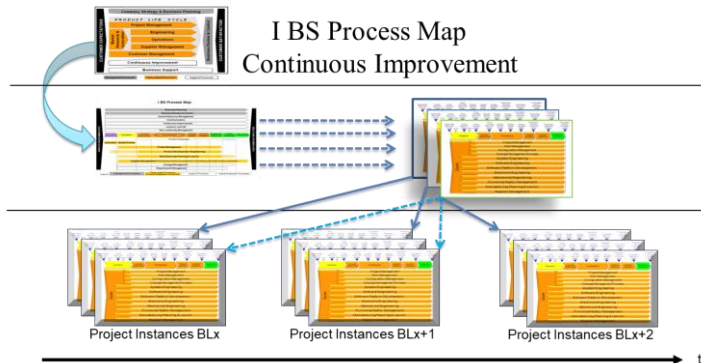
- ◆ 需要一个方案能使遍布全球的14个事业部保持过程一致
- ◆ 用户的工作必须与已经存在的过程 and 标准相符 (例如ISO26262、ASPICE)

Ω 解决方案:

- ◆ 自2007起，采用stages定义统一的、模块化的、易扩展的过程，并使过程与相关标准合规
- ◆ 12个Stages Server，约14300个License
- ◆ 与项目管理工具、PLM、ALM集成

Ω 收益:

- ◆ 所有事业部全部通过ASPICE Level 3认证





Stages 部分用户列表



Automotive

Audi
Automotive Lighting
Blaupunkt
BMW
Bosch
Brose
Continental
Daimler
Delphi
Denso
General Motors
ETAS
Hella
HERE
Knorr-Bremse
Magna
Marquardt
Panasonic
Thales
TRW
Volkswagen
Webasto
ZF
Zollner
CATL
UAES
NIO
PATAC

...



Healthcare

Alere
Braun
Carl Zeiss
Fresenius
Olympus
Siemens
Ziehm Imaging



IT/ Telecommunications

GFT
Intersoft
Nash Technologies
NEC
Micronas
PTC
Teleca



Engineering/ Automation

ABB
Carl Schenk
EBM Papst
Heidelberger
Druckmaschinen
Insta
Kratzer Automation
Mettler Toledo
Mühlbauer Group
Rohde&Schwarz
Siemens Industries
Wago



Banking

Landesbank Kiel
Fiducia
Credit Suisse
UBS



Defense/ Aerospace

Airbus/空客
Diehl
EADS
Elbit
JAXA
KID
Orbital
Raytheon

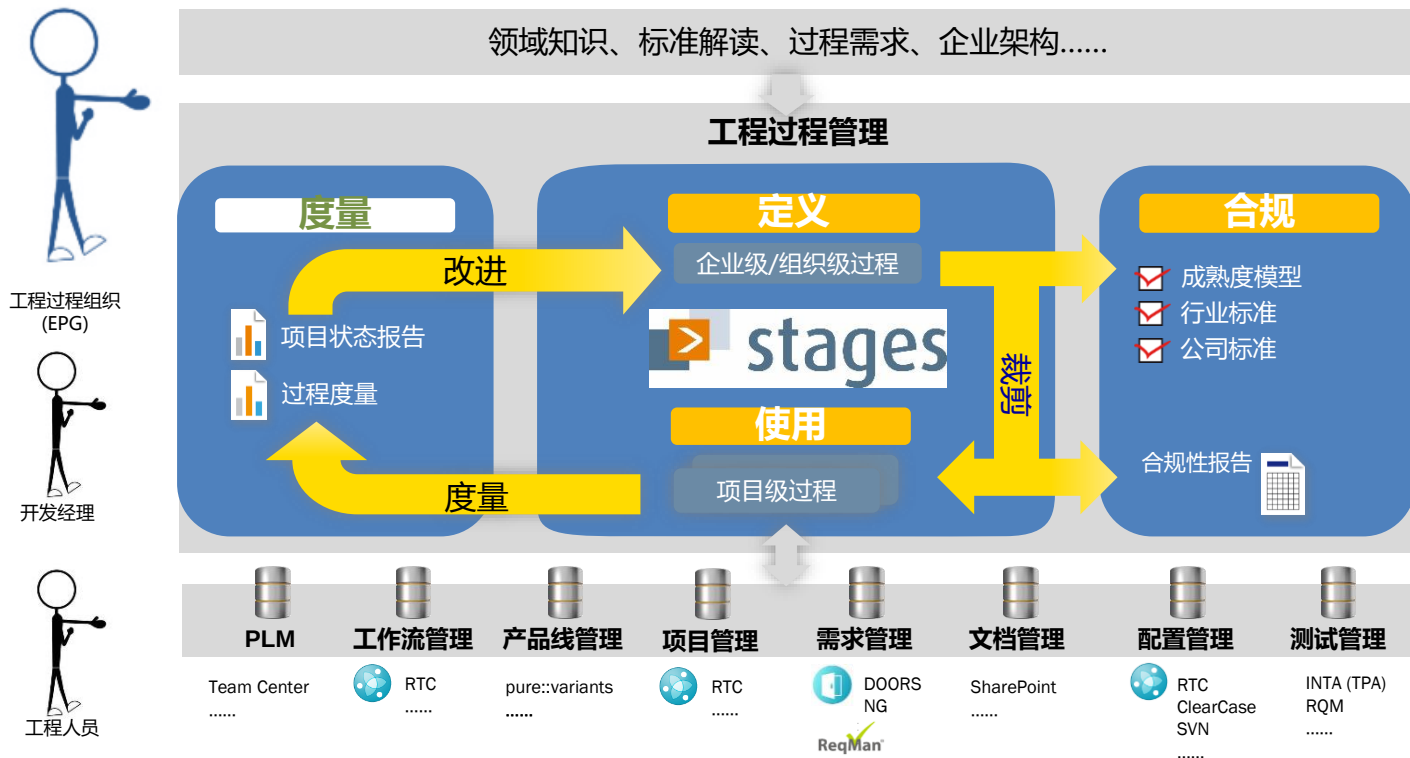
Further

Bosch und Siemens Hausgeräte
Deutsche Post
GMC Software Technologies
Raab Karcher
Giesecke & Devrient
Thales Rail Signaling
Bundesagentur für Arbeit
Kassenärztliche Vereinigung
Baden-Württemberg

PART 4

总结

Summary



感谢聆听
欢迎提问